Language Learning Product: Pedagogy and Interaction Framework

Draft based on a literature review run in Consensus (10 searches). Companion to the product idea: a language-learning app that measures what learners can do, not how often they show up.

1. The idea in one paragraph

Duolingo's main visible metric is the streak, which measures how often someone shows up, not how much they learned. Our product is built around **what the learner can recall and do**. Learners practice by producing language from memory, speaking in realistic scenarios with an AI partner, getting feedback after the conversation, and reviewing on a spaced schedule. Progress is shown as real abilities ("You can order a meal"), not a day counter. It works for complete beginners and for people who already know some of the language.

2. Who it's for

Everyone from zero to intermediate. The product adapts to the learner's level instead of forcing one experience on all of them:

- **Complete beginners** need input before output, because nobody can recall words they have never seen.
- **Learners with some knowledge** need to start producing and speaking right away.

The research does not directly test an adaptive design like this, so the adaptation rules in section 5 are a design hypothesis, not an evidence-backed claim.

3. What the research says

Product idea	Verdict	Evidence
Force recall, not recognition	Strongly supported	11 of 13 studies favored retrieval practice for vocabulary retention, none against, 2 mixed. Benefits needed about 4 or more repetitions and were larger when spaced.
Scenario / task-based practice	Supported	11 of 12 studies favored task-based teaching for speaking fluency and confidence. Traditional drills often scored higher on immediate grammar accuracy.
Judge on communication, not perfection	Partly supported	Withholding corrections during speaking protects fluency. Ignoring errors across the whole curriculum limits long-term accuracy.
AI conversation partner	Supported	Meta-analyses report medium effect sizes (g about 0.48 to 0.61) on speaking, mainly fluency, pronunciation, and anxiety reduction. Grammar gains are less reliable.
Milestones instead of streaks	Plausible, not directly tested	Rewards that give informational, mastery-based feedback support intrinsic motivation. Competitive or performance-contingent rewards undermine it.
Gamification	Supported with caveats	15 of 17 studies found learning gains, not just engagement. Gains faded as novelty wore off, and long-term testing is scarce.
Streaks	Indirect benefit only	Streaks reduce dropout and encourage regular practice, but do not build memory by themselves.

4. The core learning loop

Every session follows the same loop, with the level adjusting how much support the learner gets.

1. **Choose a scenario.** The learner picks something relevant to their life (travel, work, dating, daily life). Choice reduces the controlling feel of rewards.
2. **Learn the key phrases.** A short set of 5 to 8 useful phrases with audio and meaning. Skipped or shortened for learners who already know them.
3. **Plan.** A brief pre-task planning moment. Planning reduced hesitation and false starts in task-based studies.
4. **Recall.** The learner types or says the phrases from memory, with no word bank. Hints arrive in stages (first letter, then a clue) before the answer is shown.
5. **Converse.** The AI plays a role (waiter, colleague, receptionist). The learner speaks or types freely and is not interrupted mid-conversation.
6. **Review after the conversation.** The AI shows the 2 or 3 most important errors using prompts first, then the correct form.
7. **Spaced review.** Missed words and phrases return over following days at growing intervals, with the direction of recall varied.
8. **Milestone.** When a scenario is completed without help, the learner earns a can-do statement backed by evidence from the session.

Note: the research found covert (silent) and spoken retrieval equally effective for memory. Forcing recall from memory is what matters, so typing is a valid option for learners who are not ready to speak aloud.

5. Adapting to the learner's level

A short placement check at onboarding asks the learner to type or say a few things from memory, not pick from options. The result sets a starting level, which the product keeps updating from performance. *These rules are a hypothesis to test.*

	Beginner	Some knowledge
Learn step	Full phrase set with audio	Short or skipped
Recall step	Staged hints available	Minimal hints

	Beginner	Some knowledge
Conversation	Guided, with phrase suggestions on request	Open-ended
Feedback style	Prompts with more explicit help	Prompts first, explanation if needed
Scenario difficulty	Short, predictable exchanges	Longer, less predictable

Support fades as the learner improves. The research notes that conversational bots become less useful for more proficient speakers when responses turn repetitive, so advanced scenarios should vary more.

6. Feedback design

This is the area where the evidence changed our original plan the most.

- **Do not interrupt mid-conversation.** Immediate correction during continuous speaking tended to hurt fluency and raise anxiety. Delayed or post-task feedback protected fluency.
- **Do not skip correction entirely.** Learners whose errors were ignored throughout performed worse on delayed accuracy than those who received prompts or recasts.
- **Prompts before direct answers.** Prompts (for example, "which tense fits here?") produced higher uptake and self-repair than recasts, especially for lower-proficiency learners.
- **Graduated hints, not one-click rewrites.** Research on LLM feedback warns that instant rewrites can replace the effort that produces learning.
- **Prioritize high-yield errors.** Tense and word order respond well to feedback. Articles and prepositions are more persistent. Show only the 2 or 3 most important errors per session.
- **Timing is not one-size-fits-all.** The best timing depended on working memory and anxiety. Delayed feedback is the safer default, and this is worth testing per learner.

7. LLM feedback rules

LLMs gave high-precision feedback on surface errors, matching or slightly exceeding human teachers in several studies, but they overcorrect and are less reliable on discourse and on non-English target languages.

- Use few-shot examples and domain-specific prompts. One study raised error recall from 10% to 55% at 94% precision this way.
- Add an overcorrection guard. Do not flag acceptable or idiomatic phrasing as wrong, and prefer minimal edits over rewriting.
- Prioritize errors by importance before showing any.
- Deliver graduated hints before the answer.
- Test target languages separately. Accuracy was weaker for morphologically complex languages in the studies reviewed.
- Test speech input specifically. Most LLM accuracy studies used written text, and speech recognition errors can cause false corrections.

8. Measuring progress

The headline metric is **what the learner can do**, not how many days in a row they showed up.

- **Recall strength:** how many phrases the learner can produce from memory, tracked over time.
- **Can-do list:** scenarios completed unaided.
- **Error trends:** which mistakes are shrinking over time.
- **Response time:** how quickly the learner produces language, as a fluency signal.

9. Motivation design

- **Keep a light habit mechanism.** Streaks and reminders reduce dropout. Show them as a secondary element, not the headline.
- **Start with completion, move to mastery.** Completion rewards help people begin. Mastery-based milestones sustain motivation.
- **Make rewards informational.** A milestone should tell the learner what they can now do, and not feel like a grade.

- **Avoid leaderboards and monetary rewards.** Competition energized some learners and demotivated others, and large extrinsic incentives shifted attention to points over communication.
- **Give choice.** Letting learners pick scenarios reduces the controlling effect of rewards.

10. Positioning

Duolingo builds the habit of showing up. This product builds the ability to speak. It can be pitched as the next step after gamified apps, or as a standalone product for anyone who wants real-world speaking ability.

11. Limits of the evidence

- Most studies are classroom studies with university EFL learners, often short in duration. Results may not transfer to a self-paced app, other languages, or other age groups.
- The reviews on ignoring minor errors and on streaks had too few directly relevant papers for a Consensus Meter reading. Treat them as weaker evidence.
- No study directly tested whether milestone-based rewards prevent churn better than streaks.
- Chatbot gains were mainly in fluency and vocabulary, with less reliable gains in grammar.
- Consensus summaries are AI-generated. Open the underlying papers before relying on any key claim.

12. What to test first

1. **Speech vs. text feedback accuracy.** Does the LLM correct spoken input reliably after transcription?
2. **Milestones vs. streaks.** Does a milestone-led design keep people coming back?
3. **Level adaptation.** Does the placement check set a sensible starting level, and does fading support work?
4. **Feedback timing.** Is delayed feedback the best default for most learners?
5. **Retention over months.** Do learners still recall phrases after 30 days or more?

13. Suggested first version

- One target language and one learner background, so results are easier to interpret.
- Three to five scenarios.
- Placement check, the learning loop in section 4, and spaced review.
- Post-conversation feedback with the LLM rules in section 7.
- A can-do progress screen and a light reminder system.

14. Open questions

- Which language and learner background should come first?
- Voice-first, text-first, or both? Both are supported by the evidence, but voice adds speech recognition errors.
- How much input content does a beginner need before recall tasks make sense?
- How should the product handle learners who want accuracy drills, given that traditional drills scored higher on immediate grammar accuracy?